

These observations support the use of a von Mises–Fisher (vMF) model as a simple and effective approximation for the directional geometry, with a shared concentration parameter κ .

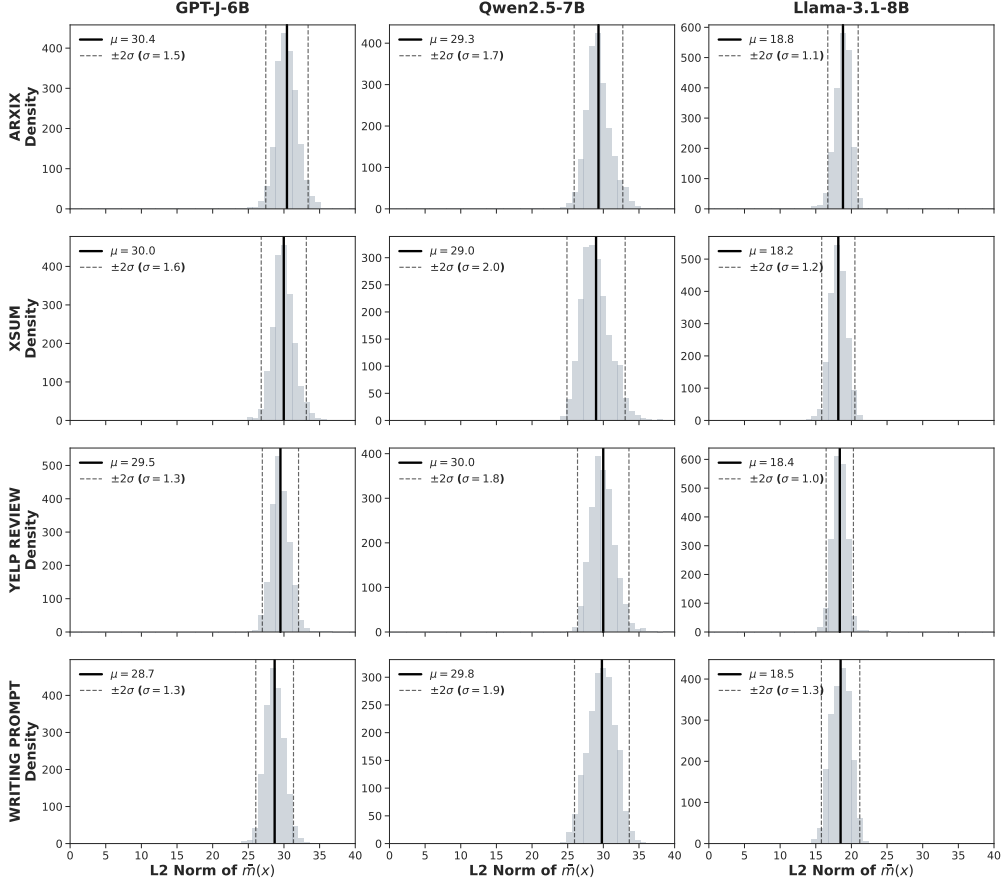


Figure 5: Empirical distributions of L_2 norms of representations obtained from the last 8 layers and the final 20% of tokens, across different models and domains. Columns correspond to *EleutherAI/GPT-J-6B*, *Qwen/Qwen2.5-7B*, and *meta-llama/Llama-3.1-8B*, while rows correspond to the Arxiv, XSum, Yelp, and Writing datasets. The solid and dashed red lines denote the mean and the $\pm 2\sigma$ intervals, respectively, with exact values reported in the upper-left legends.

F Additional Experiment results

F.1 Score Distributions

In this section, we visualize score distributions from four LLMs under different settings. We use *meta-llama/Llama-3.1-8B* as the observer model. The training data is generated by *GPT-3.5-Turbo*. The evaluated datasets are generated by *GPT-3.5-Turbo*, *Claude-Instant*, *Google-PaLM*, and *Llama-2-70B*.

In Figure 7, the leftmost column shows scores from RepreGuard [33], the middle column corresponds to raw representations without steering, and the rightmost column shows scores produced by our method. Although the first two approaches exploit representation-level separability, a non-negligible overlap between the two classes remains, which limits detection performance. This issue is particularly pronounced under OOD settings: RepreGuard exhibits 15.24% and 17.25% overlap on datasets generated by *Claude-Instant* and *Google-PaLM*, respectively, while S2D without steering yields 14.10% and 18.27%. In contrast, our method achieves substantially clearer separation between human-written and LLM-generated text, even under OOD settings, leading to improved discriminability.

F.2 Large-Sample Validation of Low-FPR Interpolation

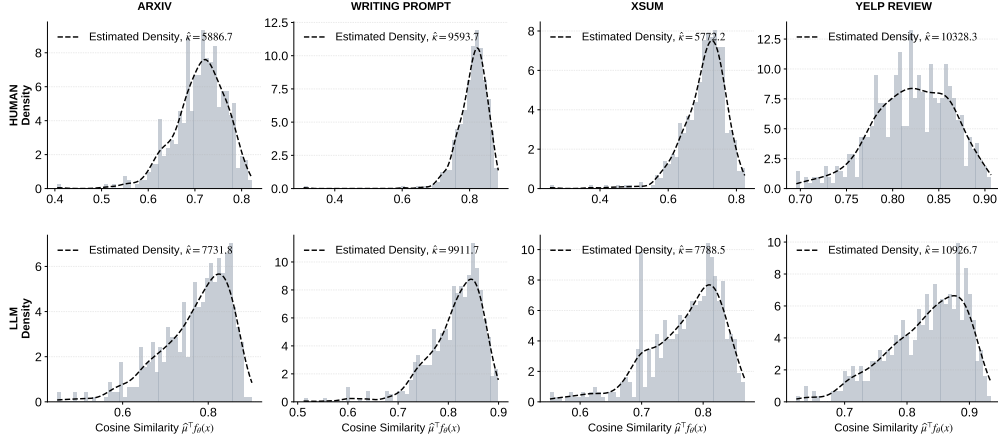


Figure 6: Empirical distributions of projected representations $\hat{\mu}^\top f_\theta(x)$ across domains. Histograms show empirical frequencies, and dashed curves represent the corresponding estimated densities. The estimated concentration parameter κ is reported for each case.

Each DetectRL test set contains 1,000 human-written samples, so the smallest nonzero empirical FPR is 0.1%. Therefore, TPR@0.01% in the main experiments is obtained by linear interpolation of the empirical ROC curve. We conduct an additional large-sample experiment to examine the effect of this interpolation. Specifically, we collect 21686 human-written texts and 20,000 LLM-generated texts from C-ReD, EvoBench, and OpAI-Bench. The resulting empirical FPR resolution is approximately $1/21686 \approx 0.0046\%$, allowing us to evaluate performance directly near FPR = 0.01%. We compare the interpolated and directly evaluated TPR@0.01% for S2D, RepreGuard, and ImBD. As shown in Table 7, the interpolated and direct estimates differ by at most 0.09 percentage points across the three methods. This suggests that, in this larger-sample evaluation, linear interpolation has little effect on the reported TPR@0.01% values.

Table 7: Large-sample validation of TPR@0.01%. All values are percentages.

Method	Interp.	Direct	Gap
S2D	38.61	38.52	0.09
RepreGuard	13.99	13.99	0.00
ImBD	11.12	11.10	0.02

F.3 Shots of Training Dataset

We evaluate the impact of training set size on detection performance. The training sets are randomly sampled from a mixed-LLM dataset with sizes $\{16, 32, 64, 128, 196, 256, 384, 512\}$ pairs. We use *Llama-3.1-8B* as the base model and *Llama-3.1-8B-Instruct* as the rewriter, and evaluate performance on four test sets (*ChatGPT-3.5-Turbo*, *Claude-Instant*, *Google-PaLM*, and *Llama-2-70B*). As shown in Figure 8, S2D achieves strong data efficiency and consistent improvement over baselines, especially in low-resource settings. We observe a slight performance drop at the earliest stage as training size increases, since a very small number of samples can perturb the estimated score distribution. Nevertheless, even with as few as 16 pairs, S2D can still exploit the inherent score gap between human-written and LLM-generated texts. As more training data becomes available, the estimated distribution stabilizes, resulting in clearer separation and improved performance.

F.4 Long-Context Steering Stability

Since the same steering vector is injected at every valid token, we examine whether long sequences cause representation saturation or collapse. We train S2D on 512 mixed DetectRL human-LLM pairs using *Meta-Llama-3-8B* and evaluate a fixed steering checkpoint on a separate long-context diagnostic set constructed from OpAI-Bench, C-ReD, and EvoBench, with 400 human-LLM pairs at each sequence length. As shown in Table 8, performance improves with

Table 8: Long-context stability of S2D.

Tokens	AUROC	TPR@1%	Overlap
256	0.9107	0.610	0.105
512	0.9667	0.790	0.096
1024	0.9940	0.960	0.055
2048	0.9998	0.990	0.035
4096	0.9999	0.990	0.036

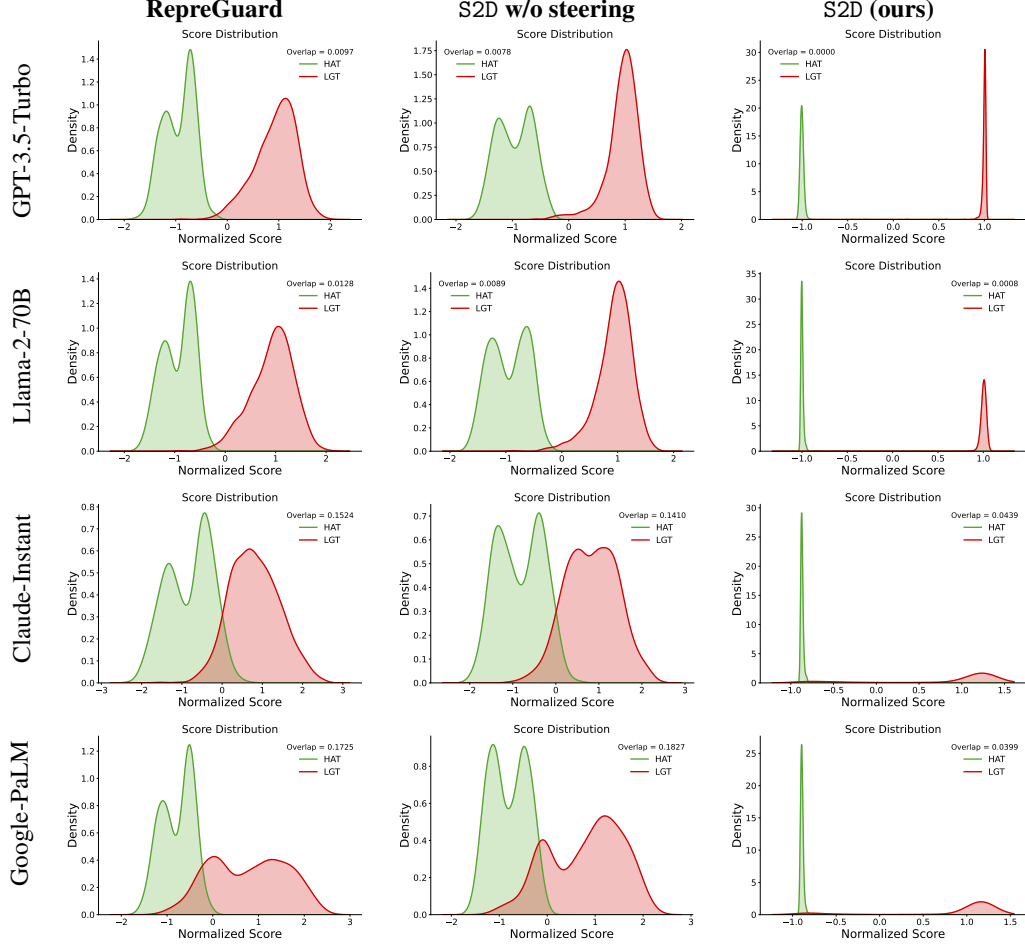


Figure 7: Score distributions of different detection methods involving hidden representations across datasets. Columns denote different methods and rows denote datasets generated by different LLMs. HAT and LGT denote human-authored text and LLM-generated text, respectively.

context length and plateaus beyond 2048 tokens, while score overlap decreases from 0.105 to about 0.035. We therefore observe no evidence of length-induced representation collapse up to 4,096 tokens. This result is an empirical diagnostic rather than a guarantee for arbitrarily long contexts.

F.5 Different Observers

Table 9 shows that strong observer models (e.g., *Llama-3.1-8B*, *Qwen2.5-7B*, and *Falcon-7B*) achieve consistently near-saturated performance across all generators, with high AUROC and stable TPR even at very low false positive rates. In contrast, weaker models (e.g., *Mistral-7B-v0.3*, *GPT-Neo-2.7B*, and *Gemma-2-9B*) exhibit both lower accuracy and larger variance across generators, suggesting sensitivity to distributional shifts and less discriminative features. Notably, instruction-tuned variants do not consistently improve performance, suggesting that detection effectiveness in S2D is primarily determined by representation quality, rather than model scale or alignment alone.

Observer Model↓	ChatGPT			Llama-2-70B			Google-PaLM			Claude-Instant			Avg.		
	AUROC	TPR@1%	TPR@0.01%	AUROC	TPR@1%	TPR@0.01%	AUROC	TPR@1%	TPR@0.01%	AUROC	TPR@1%	TPR@0.01%	AUROC	TPR@1%	TPR@0.01%
Llama-3.1-8B	99.99	99.90	99.80	98.95	98.90	98.10	99.83	98.50	97.40	99.70	98.90	96.60	99.62	99.05	97.98
Mistral-7B-v0.3	68.85	29.20	24.20	67.55	33.80	26.60	67.97	35.40	28.40	69.72	23.70	13.90	68.52	30.53	23.28
GPT-Neo-2.7B	84.44	33.70	8.30	79.97	41.50	19.40	79.63	20.40	6.70	85.46	35.00	10.90	82.38	32.65	11.33
OPT-2.7B	99.95	99.80	82.60	99.93	99.30	94.10	99.66	95.80	74.50	99.88	98.30	86.80	99.86	98.30	84.50
Qwen2.5-7B	99.99	99.90	99.80	99.99	99.90	98.90	99.73	98.60	96.20	99.99	99.90	99.70	99.93	99.58	98.65
Falcon-7B	99.99	99.90	99.80	99.99	99.90	99.60	99.89	98.10	95.70	99.98	99.70	98.00	99.96	99.40	98.28
Falcon-7B-Instruct	99.98	99.90	99.60	99.99	99.90	95.70	99.89	97.00	93.20	99.97	99.90	95.10	99.96	99.18	95.90
Gemma-2-9B	88.01	25.70	5.90	76.25	13.90	0.10	56.10	20.90	5.20	85.79	16.50	1.00	76.54	19.25	3.05
Gemma-2-9B-Instruct	96.25	62.70	32.50	95.66	60.10	29.70	54.10	15.60	7.30	70.34	12.50	6.10	79.09	37.73	18.90

Table 9: Impact of observer model. Detection performance of S2D across various observer models.